

Title

Development and validation of an Iron–Folic Acid Supplementation Counseling Quality Assessment Tool during routine antenatal care

Research question

Can a standardized, patient-centered instrument be developed and validated to assess the quality of IFAS counseling delivered during routine antenatal care?

Null Hypothesis (H_0)

There is no statistically significant association between the quality of iron–folic acid supplementation (IFAS) counseling and adherence to iron–folic acid supplementation among pregnant women attending routine antenatal care.

Abstract

Pregnancy-related anemia remains a major public health problem, particularly in low- and middle-income countries, despite the availability of effective preventive interventions. Daily iron–folic acid supplementation (IFAS) is recommended during pregnancy, but its real-world impact is limited by poor adherence and inconsistent counseling during antenatal care. This study aims to develop and validate a standardized patient-reported tool for assessing the quality of IFAS counseling during routine antenatal care. The study will be conducted in two phases: item generation and expert content validation, followed by cross-sectional psychometric validation among pregnant women attending antenatal care. The final tool is expected to improve measurement of counseling quality, identify gaps in current practice, and support future quality improvement and implementation research.

Introduction/Review

Background

Pregnancy-related anemia remains a major public health problem, especially in low- and middle-income countries, despite the availability of effective preventive strategies. Daily iron–folic acid supplementation (IFAS) is a recommended intervention, but its real-world impact is limited by poor adherence and inconsistent implementation during routine antenatal care. Evidence from the literature shows that counseling quality strongly influences adherence, yet counseling is often incomplete, non-standardized, and variably delivered across clinical settings. Several counseling quality instruments have been developed in other healthcare contexts, especially family planning and specialized perinatal counseling. However, these tools were not designed for IFAS counseling during routine antenatal care and do not comprehensively assess

the technical, behavioral, and patient-centered components that are relevant to maternal nutrition. This creates a methodological gap and supports the need for a standardized, validated tool specifically for IFAS counseling.

Rationale

The study is needed because IFAS counseling is a modifiable determinant of adherence, but there is currently no standardized patient-centered instrument that comprehensively measures the quality of IFAS counseling in routine antenatal care. Developing such a tool would allow better assessment of counseling practice, improve comparability across studies, support quality improvement, and strengthen future implementation research.

Aim

To develop and validate a standardized Iron–Folic Acid Supplementation Counseling Quality Assessment Tool for routine antenatal care.

Objectives

Primary objectives

- To generate an initial pool of IFAS counseling items from the literature and guidelines.
- To assess the content validity and face validity of the draft tool.
- To evaluate the internal consistency and reliability of the final tool.
- To assess the construct validity of the tool.

Secondary objectives

- To explore the association between IFAS counseling quality and IFAS adherence.
- To identify the most important domains of IFAS counseling during routine antenatal care.

Methodology

Study design

This will be a **methodological study** with two phases. Phase 1 will involve item generation and expert validation. Phase 2 will be a **cross-sectional validation study** among pregnant women attending antenatal care. If feasible, a small subset will be reassessed for test-retest reliability.

Study setting

The study will be conducted in the antenatal care clinic of **Demerdash Hospital, Egypt**.

Study period

The study will be conducted over **6 months**.

Participants

The study population will be **pregnant women attending routine antenatal care and receiving IFAS**.

Inclusion criteria

- Pregnant women aged 18 years or older
- Receiving routine antenatal care
- Prescribed IFAS
- Able and willing to provide informed consent

Exclusion criteria

- Women with severe illness preventing interview
- Women unable to communicate adequately
- Women who decline participation
- Women not receiving IFAS

Sampling method

A **consecutive sampling** technique will be used. All eligible pregnant women attending the antenatal clinic during the study period will be invited to participate until the required sample size is reached.

Sample size

The sample size will be calculated using **OpenEpi version X.X** based on the formula for comparing two proportions. The calculation will be performed using a two-sided confidence level of **95% ($\alpha = 0.05$)** and a study power of **80% ($\beta = 0.20$)**. The baseline proportion (P1) and the expected proportion (P2) will be obtained from the most relevant published literature on adherence to iron–folic acid supplementation and counseling quality before finalizing the protocol. The calculated sample size will be increased by **10%** to compensate for possible non-response or incomplete data.

Data collection tool

The data collection form will include:

- Socio-demographic data
- Obstetric history
- ANC characteristics
- IFAS counseling quality items

- IFAS adherence items
- Optional knowledge questions

The counseling quality tool will be scored simply, for example:

- **0 = not done**
- **1 = partially done**
- **2 = fully done**

A total score will be calculated, and higher scores will indicate better counseling quality.

Study phases

Phase 1: Item generation

An initial pool of items will be developed from:

- WHO recommendations
- existing IFAS studies
- counseling quality literature
- validated counseling tools from other fields

The draft domains will likely include:

- anemia education
- importance of iron and folic acid
- dose, timing, and duration
- missed dose advice
- side-effect counseling
- side-effect management
- dietary interactions
- adherence reinforcement
- patient understanding
- opportunity for questions
- family support where applicable

Phase 2: Content validity

An expert panel of obstetricians, maternal health researchers, and clinicians experienced in antenatal care will review the draft tool for relevance, clarity, and completeness. Items will be revised or removed based on expert feedback, and a content validity index will be calculated.

Phase 3: Face validity and pilot testing

The draft tool will be piloted on a small group of pregnant women to assess clarity, wording, acceptability, and completion time. The wording and response options will be refined based on pilot feedback.

Phase 4: Validation

The final tool will be administered to the main validation sample. Psychometric testing will include:

- internal consistency
- item-total correlation
- exploratory factor analysis
- construct validity
- test-retest reliability in a subset if feasible

Variables

The primary outcome variable will be the IFAS counseling quality score, measured using the newly developed questionnaire. Tool validity will be evaluated through content validity, face validity, construct validity, and item-total correlation, while reliability will be assessed using Cronbach's alpha and, where feasible, test-retest reliability.

The secondary outcome variable will be IFAS adherence, defined as consumption of at least 65% of the prescribed iron–folic acid tablets during the assessment period.

Predictor variables will include maternal age, educational level, occupation, residence, gestational age, gravidity, parity, number of antenatal care visits, history of anemia, previous IFAS use, and duration of supplementation. Potential confounding variables such as maternal education, socioeconomic status, gestational age, and ANC attendance will be considered during statistical analysis.

Data sources and measurement

Data will be collected immediately after the antenatal consultation to reduce recall bias. Trained data collectors will administer the structured questionnaire in a private setting. The patient-reported tool will be used to measure counseling quality, and adherence will be measured using a predefined cutoff.

Bias

Immediate post-consultation interviewing will reduce recall bias. Consecutive sampling will reduce selection bias. Pilot testing will help reduce measurement error. Standardized instructions for data collectors will reduce interviewer bias.

Quantitative variables

The counseling quality score will be treated as a continuous variable and, if needed, categorized into levels such as poor, fair, and good. IFAS adherence will be treated as a binary variable using a predefined cutoff. Sociodemographic and obstetric variables will also be analyzed as appropriate.

Statistical methods

Descriptive statistics will summarize participant characteristics and counseling quality scores. For validation:

- Cronbach's alpha will assess internal consistency
- Exploratory factor analysis will assess construct structure
- Item-total correlation will assess item performance
- Test-retest reliability will be assessed if feasible

For the secondary objective:

- The association between counseling quality and IFAS adherence will be tested using chi-square tests, t-tests, correlation analysis, or logistic regression depending on the data distribution.

A p-value of ≤ 0.05 will be considered statistically significant.

Ethical considerations

Ethical approval will be obtained from the relevant institutional review board before starting the study. Written informed consent will be obtained from all participants. Participation will be voluntary, and refusal will not affect the care received. Confidentiality will be maintained by anonymizing the data and restricting access to the research team.